Exp No:5 Experiment to detect outliers in a given data set.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

array = np.array([10, 15, 19, 22, 25, 28, 31, 34, 37, 39, 42, 44, 47, 49,
110, 140])

print("Array:")
print(array)

print("Mean:")
print(array.mean())

print("25th Percentile:")
print(np.percentile(array, 25))

print("50th Percentile:")
print(np.percentile(array, 50))

print("75th Percentile:")
print(np.percentile(array, 75))

print("100th percentile:")
```

```
print(np.percentile(array, 100))

def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
print("Lower Range:")
print(lr)
print("Upper Range:")
print(ur)

plt.figure(figsize=(5, 2.5))
plt.hist(array, bins=5)
plt.show()

new_array = array[(array > lr) & (array < ur)]
print("Array after removing outliers:")
print(new_array)

plt.figure(figsize=(5, 2.5))
```

```
plt.hist(new_array, bins=5)
plt.show()

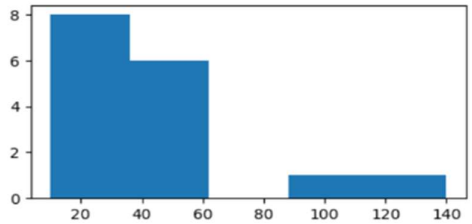
Ir1, ur1 = outDetection(new_array)
print("New Lower Range:")
print(Ir1)
print("New Upper Range:")
print(ur1)

final_array = new_array[(new_array > Ir1) & (new_array < ur1)]
print("Final Array:")
print(final_array)

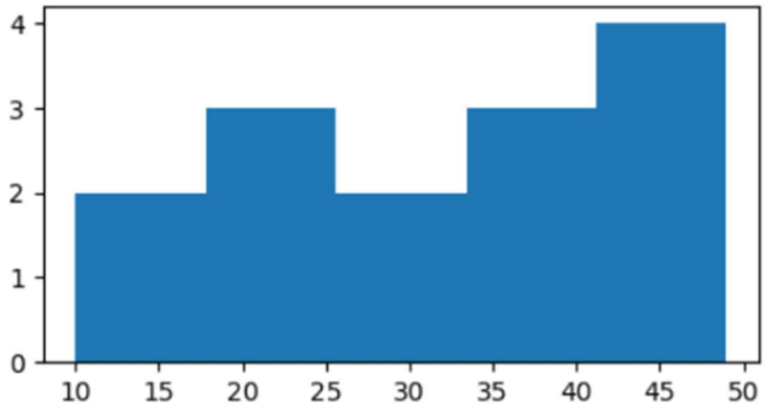
plt.figure(figsize=(5, 2.5))
plt.hist(final_array, bins=5)
plt.show()
```

Output:

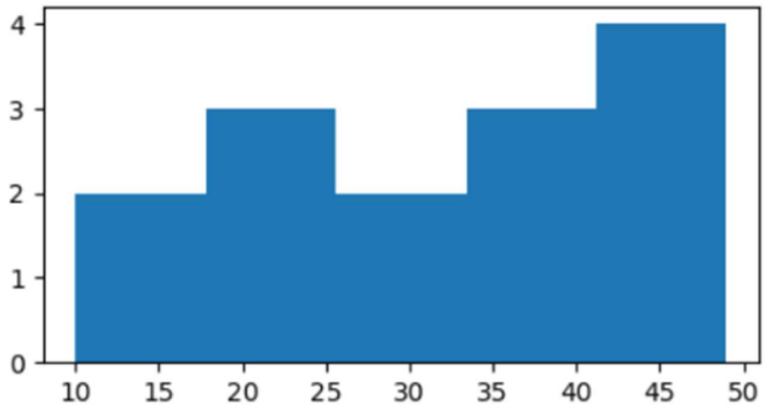
```
Array:
[ 10  15  19  22  25  28  31  34  37  39  42  44  47  49 110 140]
Mean:
43.25
25th Percentile:
24.25
50th Percentile:
35.5
75th Percentile:
44.75
100th Percentile:
140.0
Lower Range:
-6.5
Upper Range:
75.5
```



```
Array After Removing Outliers:
[10 15 19 22 25 28 31 34 37 39 42 44 47 49]
```




```
New Lower Range:
-5.0
New Upper Range:
69.0
Final Array:
[10 15 19 22 25 28 31 34 37 39 42 44 47 49]
```



Result:

Hence, the procedure to identify the outliers in a given dataset is analysed.